

Dissent Within a Transparent Central Bank: Causes and Effects *

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Abstract

How does dissent feature in a central bank with fully transparent deliberations? By studying a uniquely transparent central bank, the Swedish Riksbank, I study in detail the causes and consequences of dissent in central bank communication. First, I map dissent to governor demographics and highlight some characteristics co-varying strongly with general dissent, for example, having a higher academic degree. Second, I show that a one-SD increase in general dissent reduces short term interest rate effects by roughly a third compared to the mean. Furthermore, studying the effects of the direction of dissent, I observe that dissenting in the dovish direction drives both shorter and longer rates upwards. Finally, when studying impulse responses from monetary policy shocks induced by transcript releases in different dissent regimes, I see evidence that dissent has an effect on the shape of the subsequent interest rate path. **Keywords:** *central bank communication; dissent; monetary policy; text analysis; high frequency outcomes.*

1 Introduction

For a long time, the procedures of a central bank were something hidden and even mysterious to the public. The former chair of the Federal Reserve, Alan Greenspan, took pride in being able to, with his so called Fedspeak, perform his "mumbling with great incoherence". Since then, central banks have broadly adopted transparent communication frameworks. A transparent communication is believed to make the transmission of monetary policy more effective since it clarifies to all agents in the economy what is going on.

One central bank that has become notably transparent is Sveriges Riksbank, the Swedish central bank. When announcing a monetary-policy decision, the Riksbank publishes a Monetary Policy Report that includes the executive board's forecast for the repo rate three years ahead. The accompanying press release also reports whether any board members have entered reservations against the decision or the forecast. Around ten days

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later, the Riksbank publishes the transcripts of the meeting where the decision on the interest rate was made, which include an attributed transcript of each board member's contributions to the monetary-policy discussion. Using this uniquely transparent nature of the Riksbank, one can answer vital questions concerning the effects of transparency in central bank communication.

The contributions of this paper are threefold. First, I develop a database that includes all statements from meetings of the executive board by individual governors from 2007 until today. This database consists also of voting records, as well as literal statements, but first and foremost a sentiment score for each governor at each meeting. I build this sentiment score by employing an off-the-shelf LLM with which I classify the statements of the governors in a hawkish/dovish dimension to get at each governor's preference for monetary policy in each period. Using this sentiment variable, I create a new variable to measure dissent that captures dissent in a much more nuanced way than the discrete forms of dissent that can, for example, be read off voting records.

Second, I use this database to examine if and how dissent in a central bank affects financial markets as well as real macro economic variables. When protocols are released, financial markets usually react to the content of these protocols. I document a novel fact that the size of financial market outcomes shrinks when there is more dissent on the board. The size of this effect is large and reduces the size of the shocks by about 33%.

Third, I also perform a dissent state decomposition of protocol releases and thereafter perform state-dependent Jordà projections, obtaining the result that interest paths seem to behave differently in different interest rate regimes.

Related literature. The literature on central bank communication is a large and growing one. Among the earlier contributions on quantifying the text contents of central bank transcripts is for example Apel and Blix Grimaldi, 2014 who use a dictionary approach. By building an index of a simple word count, they show that this can predict future repo rate decisions, a feature also of the index in this paper. A more recent paper by Apel et al., 2022 uses the same dictionary approach as the paper by Blix Grimaldi and Apel, but also introduces a similarity measure to assess committee agreement. There they show that unity is higher preceding periods of monetary tightening, showing support for distaste to raise interest rates until there is internal consensus.

With the advent of large language models, there has been quite some advances in the analysis of textual information. A recent study from the ECB, Geiger et al., 2025, develops a Hawk-O-Meter to study the policy attitude of the ECB over time. This is an aggregate index that does not feature individual governor statements since the ECB maintains discretion in their statements. Even more recent work on this can be seen in Howes et al., 2026 and Baird et al., 2026. Howes et al., 2026 construct a data set of FOMC meeting transcripts to study the individual monetary policy preferences. Their comment on dissent is that it might curtail the influence of dissenters in following meetings. However rich their data set, they do not use it to draw inferences about aggregate economic outcomes beyond the central bank itself. Baird et al., 2026 have recently provided a database of central bank transcripts, but it contains neither press releases (that are released in conjunction with interest rate decision) nor governor specific statements. Furthermore, the analysis done in this paper is also not possible conduct for other central banks since the structure of the Riksbank with its scheduled transcript releases is unique and phenomenal for identification.

Modeling of central bank communication features in papers like Meade and Stasavage, 2008, Riboni and Ruge-Murcia, 2014, Blot et al., 2025 where we see examples of models trying to rationalize among other things, dissenting behaviour within a central bank board. In Meade and Stasavage, 2008, making transcripts public seems to have made FOMC governors less dissenting, in line with model predictions highlighting the risk of showing incompetence by dissenting and being incorrect. Career concerns then motivate sticking to the herd. Riboni and Ruge-Murcia, 2014 is one of few papers that study individual governor data, obtained from Riksbank data. However, in their analysis they only use individual voting data and not a textual analysis on top of it, as in this paper, to obtain a more fine grained notion of dissenting. They show results and model predictions, that dissent votes from certain governors can explain future changes in CB policy. In Blot et al., 2025 they link dissent to a negative effect on stock prices around status quo decisions.

Moreover, Gerlach-Kristen, 2004 provides early evidence on the relevance of the information contents of monetary policy transcripts. I add to this literature by showing that short-term rates are affected differentially depending on the dissent regime

Layout of the paper: section 2 presents the data set, section 3 describes the method and some descriptives on the data set, section 4 presents the results and section 5 concludes.

2 Data

I have compiled an extensive data set on the communication and decision making of the Riksbank's executive board (EB). This has been possible due to the transparent nature of the Riksbank. The data set stands on three legs: EB meeting transcripts, press releases after EB meetings and voting records on monetary policy of the EB. Furthermore, I also have a range of financial market outcomes as macro economic outcomes.

2.1 Transcripts

The written transcripts from the meetings of the EB of the Riksbank have been collected from the Riksbank's website as well as from the Riksbank's archive website. These transcripts range from 1999-01-04 until 2025-09-01. However, in the database I include only transcripts from 2003-02-06 and onwards. The reason for this is that the transcripts from that date and onwards are structured in a similar fashion. There has been one major change in the composition of these transcripts over time. From 2007-06-19 the board deanonymized the statements¹ in the transcripts, making it possible to see exactly who said what during the meetings. This structure has remained until today. In total this gives us 140 transcripts. A unique feature of the Riksbank, is that the transcripts are released at fixed day and time after each meeting of the EB. The timing is decided upon at each meeting of the board and is usually set to 09:30 one or two weeks after the meeting. A definitional note on the transcripts: I will throughout this paper use the term "transcript" over "minute". This is because of the nature of the monetary policy communication in the Riksbank. These texts are not summaries or reductions

¹I will henceforth call the anonymized transcripts *opaque*.

of what was said during the meetings (as a minute would be). They are (as much as possible) the literal transcription of what was said during each meeting of the EB.

2.2 Press releases

From the same web sources as above, I have also collected all press releases (PR) following a meeting of the EB. The PRs are presented to the public the following work day after the EB convened. They often contain a condensed message on the state of the economy as well as the decision on the interest rate. If there has been other decisions relating to quantitative easing, these are also communicated here. The format of the PRs have, similarly to the transcripts, changed over time. However, these changes have been less structural and more varying in the informativeness dimension. Therefore, I have included PRs from 1999-01-28 until 2026-09-22 in the database. Following a decision by the EB in April 2009 (Sveriges Riksbank, 2009-04-02), it was decided that the board should reveal internal voting dissent already in the PR. In total this gives us 167 PRs. Similar to the transcripts, these press releases are also released at a fixed day and time, usually at 09:30 the day after the meeting of the EB has occurred.

2.3 Vote data of the EB

Publicly available is also time series data of the votes cast by the individual EB members from 1999-01-04 until 2026-09-22. In total there have been 199 votes held until today. The reason why there is not a perfect overlap with the amount of PRs and the number of votes is because of the informal nature of the Riksbank's proceedings. In the early 2000s, votes were cast a bit more irregularly. From 2003-02-06 and onwards, i.e. from the date which is the start date of the transcript data there have been 145 votes. The remaining irregularity is due to extra changes of the interest rate during the Great Recession and the onset of the Covid-19 pandemic. A definitional note on the composition of the EB: the EB of the Riksbank consisted of 6 members until 2022, and since then of 5 members. One of them is the chair of the EB while the others serve as deputy governors. If there is a tie, the chair is the tie breaker.

2.4 Monetary policy event study data base

Both PRs and transcripts throughout the history of the data I am using have been released on fixed dates and times. This makes it possible to identify the instantaneous effect of the information conveyed in them. This has already been done by Almerud et al., 2024, so I am using their database that was most recently updated in December 2025. The database contains percentage point changes in OIS rates for a series of different maturities (1 month to 10 years) in a tight window (-10 min to +20 min) around the communication events. It also contains data on reactions of the SEK exchange rate as well as the OMX SPI Swedish stock market index.

3 Digitization and descriptives

3.1 Developing the sentiment score

To analyze the contents of the text material, I have used an off-the-shelf large language model from Meta. The specific model I have used for this project is the `llama-3.3-70b-versatile` (Grattafiori et al., 2024). This model offers an "industry-leading performance in text and image understanding" (Meta AI, 2025). In addition, it is also a model that has been used earlier in similar work classifying central bank statements (e.g. Geiger et al., 2025). I have used the LLM to analyze the transcripts and the decisions from the meetings of the executive board sentence by sentence (for details about the prompting procedure, see Appendix A.2.1). Each sentence is then labeled as being either hawkish, dovish or neutral. For detailed instructions on how the LLM was instructed to label, see Appendix A.2.2. We should view hawkishness and dovishness to be existing in the same dimension and therefore mutually exclusive, if a sentence contains a stance on monetary policy.

To increase accuracy in the LLM classification, I let the language model keep the previous sentence as context when analyzing every next sentence after the first one in each statement. Adding more context sentences does not improve the accuracy of the classification. I also create a simple BOW algorithm to compare the sentiment score with. The LLM approach outperforms the BOW by being able to extract nuance in the text that the BOW can not.² Details on the validation and accuracy of the LLM can be found in Appendix A.2.3.

I then create, when possible³, a governor-meeting specific sentiment score variable, S_{gt} , that is defined as follows:

$$S_{gt} = \frac{\#hawk - \#dove}{\#hawk + \#dove}$$

The sentiment score, S_{gt} , can be viewed as the net hawkishness of a governor g in time t and will naturally assume a value between -1 and 1 . If all sentences are regarded as neutral, the score is set to 0 . The measure is similar to the ones defined in Apel and Blix Grimaldi, 2012 and Geiger et al., 2025. PRs have also been analyzed sentence by sentence, yielding an aggregate sentiment score for each meeting. I define this variable as S_t^p , which will never be governor specific. For more descriptive information on the underlying data set, look to Appendix A.3. Thus, if a governor speaks a lot, but utters equally many dovish as hawkish sentences, she will be given a near 0 sentiment score for that meeting.

3.2 Descriptives

In Figure 1 we can see the meeting mean sentiment score, defined as $S_t := \sum_g \frac{1}{|G_t|} S_{gt}$ where $|G_t|$ is the number of governors attending a meeting in time t . I plot the score for the transcripts and the PRs as well as the changes in the repo rate over time in the graph below (the red and blue bars). As can be seen, there is a high degree of co-variation between the indices and the rate, except during the ZLB period beginning in 2016. I also conclude

²For example the BOW algorithm incorrectly identifies a period of low dissent as having high dissent.

³From June 2007 and onwards this is possible.

that there is, as perhaps is not too surprising, a high degree of co-variation between the sentiment score of the transcripts and the PRs ($\rho \approx 0.87$). This is to be expected, since the PRs should summarize the contents of the transcripts.

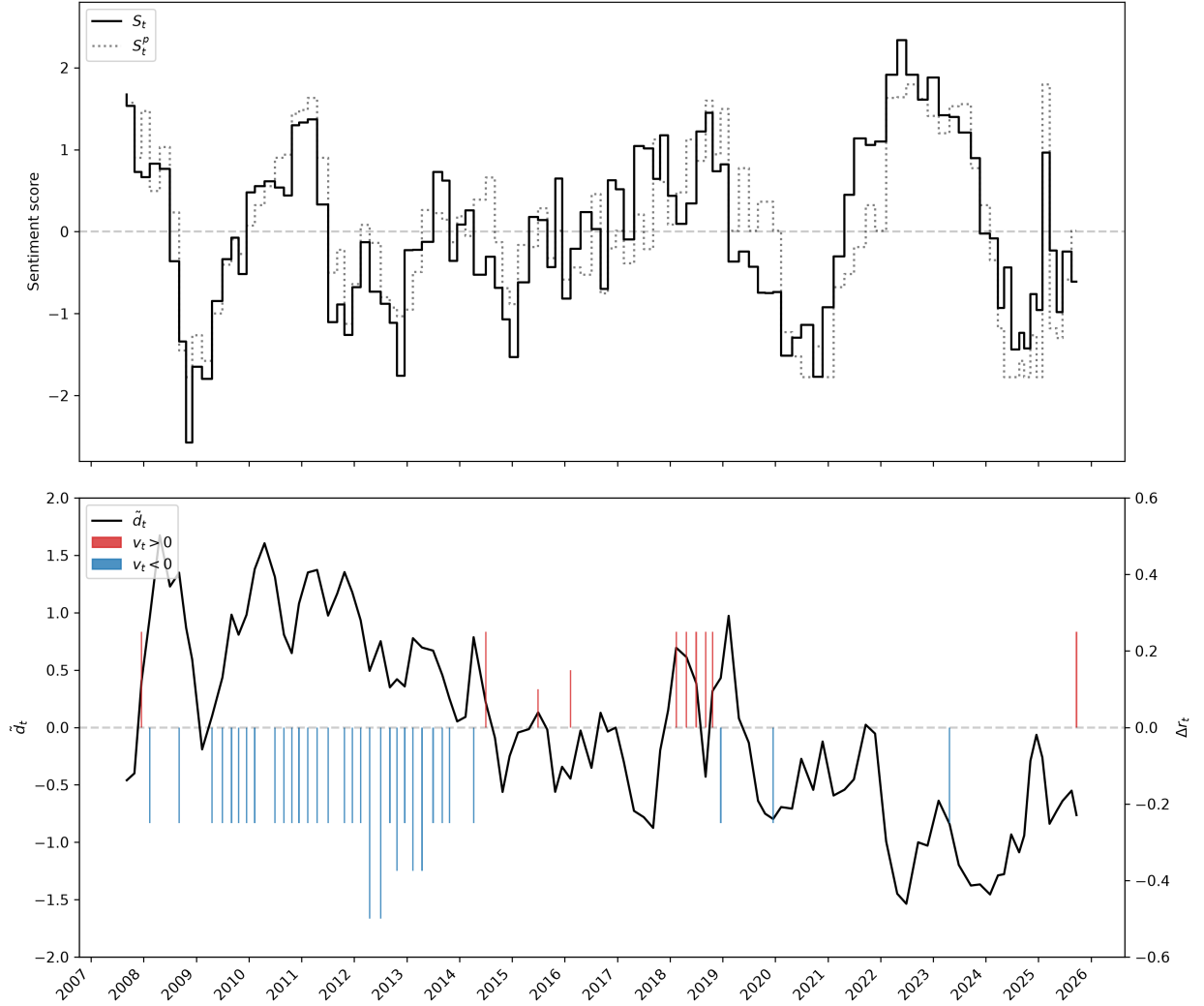


Figure 1: Mean sentiment score for transcripts and press releases over time (normalized) and repo changes, along with dissent (standardized) and vote dissent.

Since the transcripts of the Riksbank have been transparent since June 2007, I also inspect the trajectories of individual governor sentiment scores. One can thus also plot governor specific sentiment scores, which I do in Appendix A.4. Having data on individual level is what makes this data base unique and also what I will be focusing on in this paper.

3.3 A new measure of dissent

Formal dissent in the Riksbank can manifest itself in three ways. Either a member of the executive board disagrees on the interest rate, the path of the interest rate or asset purchases. These incidents of dissent are recorded in the voting transcripts of the Riksbank and binary in nature. Either you dissent on one of these three things, or you do not. However, dissent opinions in the board will not always, even seldom, materialize

in actual dissent in these variables making the usage of votes only to capture dissent in a central bank quite coarse. Therefore, the new information on sentiment that can be obtained from the data set, makes it possible to understand the nuances in dissent.

With the help of our governor specific sentiment score, S_{gt} , I define a more fine grained measure of dissent; the dispersion in terms of our sentiment score. I define governor dissent, d_{gt} , as follows:

$$d_{gt} := \left| S_{gt} - \sum_{i \neq g} \frac{S_{it}}{|G_t| - 1} \right| \quad (1)$$

which is a governor specific leave-one-out mean deviation variable. This gives us a way of measuring how far the governors are from each other in the sentiment score dimension. I furthermore define sentiment dispersion as

$$\tilde{d}_t := \frac{1}{|G_t|} \sum_g d_{gt} \quad (2)$$

where $|G_t|$ is the number of governors voting at meeting in time t . I have decided to use absolute dissent so that a meeting where there is dissent in both directions is not incorrectly misspecified as a low dissent meeting. In the lower graph of Figure 1, we plot the sentiment dispersion and overlay this with the number of dissenting votes⁴ on the interest rate in each meeting. As we can see, periods of dissent in voting of the interest rate often coincide with periods of high absolute meeting dispersion. However, there are also periods where $\tilde{d}_t$ is high but it does not materialize in voting dissent. Therefore, I argue that this measure captures dissent that was earlier unobserved.

I also define a vector valued directional dissent variable. With the help of Defs. (1) and (2) I define what I call the *latent minority*, M_t :

$$M_t := \{g_t \in G_t : d_{gt} \geq \tilde{d}_t + 1.5 \cdot \sigma_{d_t}\} \quad (3)$$

Using this definition, we define the directional dissent variable D_{gt} :

$$\delta_{gt} := \begin{cases} S_{gt} - \frac{1}{|G_t| - |M_t|} \sum_{i_t \notin M_t} S_{it}, & g_t \in M_t \\ 0, & \text{else} \end{cases} \quad (4)$$

and then for the entire latent minority as a whole

$$\Delta_t := \begin{cases} \frac{1}{|M_t|} \sum_{g_t \in M_t} D_{gt}, & M_t \neq \emptyset \\ 0, & M_t = \emptyset \end{cases} \quad (5)$$

⁴Definition of D_{gt} , direction of dissent: $D_{gt} \in \{-1, 0, 1\}$, where -1 means vote for lower rate, 0 no dissent and 1 a higher rate.

4 Results

4.1 Governor demographics

First, as a display of the great detail observed in the data, I correlate some governor characteristics, such as educational background and gender with the dissent metric d_{gt} . The dissent variable we are correlating is d_{gt} , normalized over the entire sample. I run a regression of dissent on all these variables as follows

$$d_{gt} = \alpha + \sum_i \beta_i \mathbb{1}\{\text{characteristic}_{ig}\} + \gamma \text{age}_{gt} + \delta \text{tenure}_{gt} + \epsilon_{gt}$$

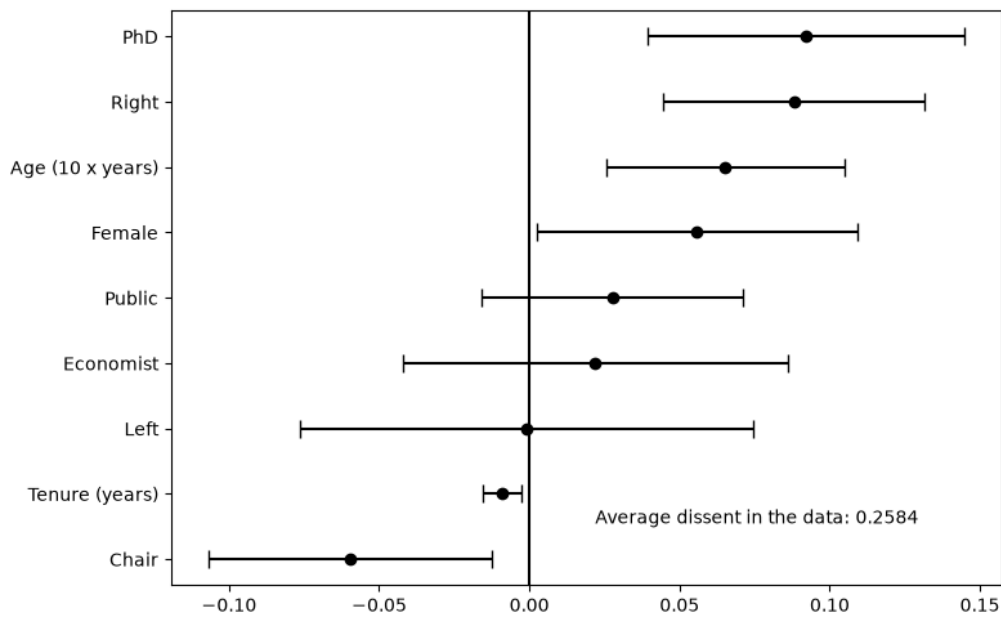


Figure 2: Absolute, standardized governor dissent correlated with governor demographics coefficient values with 95% confidence spans

The chair of the central bank is almost mechanically less dissenting, almost always being part⁵ of the deciding majority in the sample I study. By years of tenure we observe less dissent, while age as well as having a PhD seem to correlate with more dissent.⁶ In Appendix A.5, I put forward another measure of dissent that can be more closely connected to vote dissent. The results on dissent and governor demographics could be viewed from the lens of Meade and Stasavage, 2008. Career concerns could be the explanation to why older people who have an academic position tend to dissent more. We also see support for the moderating role of the chair, as in Howes et al., 2026, where we see that the chair is significantly less dissenting than other members of the board.

⁵The chair sides with the minority once in July 2014.

⁶These correlations hold also when dropping Lars E.O. Svensson, a more senior and very frequent dissenter in the data set.

4.2 Interest rates

First, I establish a correlation between the sentiment score extracted from the transcripts and the changes in the interest rate. I regress the changes in the interest rate, Δr_t , on the mean of S_{gt} ⁷. Furthermore, I also extend the regression to include S_t^p , which is the sentiment score of the related PR as well as the variable for dissent in the current and previous period and the interest rate in the previous period. I then also include controls, X_t , for inflation, unemployment and time fixed effects (year).

$$\Delta r_t = \alpha_0 + \beta_0 S_t + \beta_1 S_t^p + \beta_2 \Delta_t + \Delta r_{t-1} + \Gamma X_t + \varepsilon_t \quad (6)$$

As can be seen in Table 1, the sentiment score variable does correlate strongly with the repo rate set by the Riksbank. Controlling for the information content in the press release preceding the release of the transcripts, we see that the correlation of the sentiment score with the interest rate goes down, owing to some part to the fact that the interest rate decision already has been communicated in the PR. In Appendix A.6.1, I run the same regression with different weights for different members of the board, yielding no substantial differences.

Table 1: Results from regression (6)

	(1) Δr_t	(2) Δr_t	(3) Δr_t	(4) Δr_t	(5) Δr_t
S_t	0.214*** (0.054)	0.195*** (0.052)	0.158** (0.071)	0.155** (0.077)	0.140** (0.067)
Inflation controls		0.062** (0.025)	0.059** (0.024)	0.086*** (0.026)	0.046* (0.026)
Unemployment controls		-0.031 (0.067)	-0.020 (0.070)	0.046 (0.073)	-0.022 (0.065)
S_t^p			0.044 (0.037)	0.035 (0.042)	0.017 (0.044)
Δ_t				-0.147** (0.062)	-0.114** (0.056)
$\Delta r_{g,t-1}$					0.379*** (0.117)
Constant	-0.008 (0.016)	0.107 (0.509)	0.025 (0.533)	-0.522 (0.558)	0.079 (0.516)
R-squared	0.606	0.631	0.634	0.654	0.707
F-stat	15.515	7.440	7.761	6.210	7.606
No. obs.	140.000	140.000	140.000	113.000	113.000

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

We see in column (6) that the coefficient on dissent is positive while the coefficient on the lag is negative. This might suggest that when we are approaching the tipping point, going from increasing to decreasing rates, we might experience a final somewhat surprising increase in rates that is then followed by a decrease, when

⁷ After taking the mean on meeting level, I standardize the S_{gt} .

there is a high amount of dissent. In this regard, I would say that we by identifying periods of high dissent we also identify monetary policy tipping points. One possible explanation to the negative coefficient on the lagged dissent and positive on the current period, is that dissent could be capturing the turning point of monetary policy movements. However, given that dissent is a scalar and not a vector, we seem to be capturing only the end of interest increase cycles given the signs that we are getting.

4.3 Financial market outcomes

To estimate the causal effect of board dissent on financial markets, I exploit high-frequency changes in financial market rates within a tight window around the release of transcripts. These are changes to financial market rates that occur in a tight window (-10 min to +30 min) around the release of the transcripts. I proceed by looking at the size of market outcomes, $|Y_t|$, as a function of the aggregate deviation of the board. $|Y_t|$ is in this case long and short term interest rate swaps, as well as exchange rate reactions and stock exchange rate responses. I use the sentiment dispersion variable, defined in equation (2), and then I regress the absolute size of market outcomes on this $\tilde{d}_t$ variable. We also include controls X_t , including the level of the repo rate as well as a global VIX index as a proxy for economic risk. The VIX index is included to ensure that the results are not driven by the underlying aggregate risk state of the economy. The regression looks as follows:

$$|Y_t| = \alpha_0 + \beta_0 \tilde{d}_t + \Gamma X_t + \varepsilon_t \quad (7)$$

Results are reported in Table 2.

Table 2: Results from regression 7

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
	$ 1M $	$ 3M $	$ 6M $	$ 1Y $	$ 2Y $	$ 5Y $	$ 10Y $	$ SEK/USD $	$ SEK/EUR $
Δ_t	-0.264 (0.190)	-0.714* (0.368)	0.043 (0.084)	0.031 (0.082)	-0.003 (0.103)	0.026 (0.086)	-0.044 (0.066)	1.433 (0.897)	0.557 (0.863)
r_t	-0.587 (0.726)	-0.042 (0.525)	-0.074 (0.361)	0.011 (0.266)	-0.060 (0.230)	-0.285 (0.217)	-0.347 (0.274)	1.524 (2.840)	-1.775 (1.952)
VIX	-0.243 (0.269)	0.362 (0.422)	-0.019 (0.166)	0.197 (0.206)	-0.008 (0.182)	-0.191 (0.165)	-0.121 (0.136)	-1.670 (1.757)	-1.733 (1.239)
Constant	1.808** (0.865)	1.324** (0.625)	1.090*** (0.405)	0.545** (0.270)	0.692*** (0.262)	1.002*** (0.259)	1.063*** (0.321)	8.658*** (3.145)	10.692*** (2.388)
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
R-squared	0.351	0.233	0.381	0.380	0.275	0.329	0.288	0.164	0.278
Mean outcome size	1.14	1.281	1.007	.559	.621	.677	.668	10.42	8.694
F-stat	0.932	1.305	0.110	0.337	0.026	0.912	0.970	1.228	0.986
No. obs.	108.000	108.000	108.000	108.000	108.000	108.000	108.000	108.000	108.000

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

What can be observed is that when there is a large amount of dissent in the board, i.e. a high $\tilde{d}_t$, the size of the response on the short interest rates is substantially smaller than the average. A one standard deviation

increase in $\tilde{d}_t$ reduces the effect on the 1M interest rate by 33%. A similar effect exists also for the 3M interest rate, although smaller and non-significant. This result remains when put through a number of robustness tests (see Appendix A.8) such as dropping the most dissenting governor and dropping the 5% highest and lowest outcomes of the 1M variable. It also holds if we drop the meetings when there was dissent in voting. Importantly, using dissent in voting instead as the independent variable, yields no statistically significant effect on financial market outcomes.

What is it that drives this effect? Is it the uncertainty around the releases of the transcripts, induced by the dissent on the board, that reduces the ability of the CB to guide the market in a given direction? Due to the small amount of trade that occurs in these short term swaps, it is difficult to get numbers on volatility in the event windows. Nevertheless, I did run regressions on two different measures of volatility and did not find any significant results. See Appendix A.7 for details. Thus, this explanation can tentatively be ruled out. Still, more dispersion, as measured by $\tilde{d}_t$ seems to dampen the short-term policy response. However, the most important effect to observe here might be the 0 effect for all other financial market outcomes. A more dissenting board does not seem to make the size of financial market outcomes larger. This lends support to the Riksbank's and other central banks' decision to increase transparency in communication. There is at least nothing that overtly showing dissent has an adverse effect on markets, except for in the case of the short run.

4.4 State dependent impulse response functions

The result in Section 4.3 motivates an examination of the effects at longer horizons. To extend the analysis of the economic consequences of dissent, I perform an analysis in the vein of Jordà, 2005. I use normalized shocks to the 1 month overnight index swaps directly in the state dependent Jordà local projection

$$y_{t+h} - y_{t-1} = \alpha_h + \kappa_h \mathbb{1}\{\tilde{d}_t \geq \eta\} + \beta_h \mathbb{1}\{\tilde{d}_t \geq \eta\} \hat{i}_t + \delta_h (1 - \mathbb{1}\{\tilde{d}_t \geq \eta\}) \hat{i}_t + \sum_{l=1}^{\ell} \gamma_h X_{t-l} + \varepsilon_{t+h} \quad (8)$$

The details of this regression are as follows. The outcome variables, y , are in this case CPIFxe inflation rate, repo rate, unemployment and an index for industrial production (IPI). η is the median of the dissent variable. X_{t-l} contains the controls which include 12 lags to the outcome variables and 12 lags to the shock variable. The data set has been divided into two states, either there is high dissent or low dissent, where the splitting line is the median of dissent in the data set. I then estimate β_h and δ_h the state dependent impulse response at horizon h to a 1 SD shock to the 1M OIS swap rate for all horizons from $h = 0$ to $h = 24$. For the CPIF, unemployment and IPI the responses are not different by state and insignificant on most time horizons. However, we see that the effects on inflation and repo rate paths are significantly different and in the case of the repo rate even of different signs. This hints at the fact that dissent might arise in different stages of the monetary cycle. A decision that caused short term interest rates to go up where there was high dissent could be thought of as a moment where monetary policy was kept tight slightly longer than expected and that this later was followed by a series of interest rate decreases. This could explain the differential paths for the high and the low regime.

Going back to Figure 1, we can perhaps observe a tendency of dissent increasing when transitioning between regimes. As to why we see no effect on the other real macroeconomic variables, one can perhaps contemplate that the shocks from the press releases are tiny in magnitude and do not contribute substantially to economic outcomes in the wider economy.

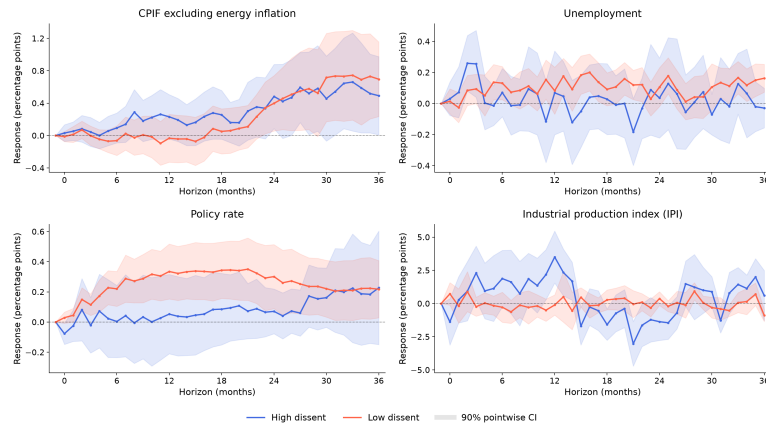


Figure 3: Impulse response functions of 4 macroeconomic variables to shocks to a 1M OIS swap rates by dissent regime.

5 Conclusion

In this paper I have provided a novel and unique data set, employing data from a uniquely transparent central bank, the Riksbank, to study the effects of transparency in monetary policy. By employing an off-the-shelf LLM and analyzing the transcripts from the meetings of the executive board, new insights on central bank communication have been obtained. I provide a new quantification of the short run effects of dissent in central bank communication. At the release of the transcripts, short-term financial market swap rate responses are dampened by around 33% when there is high dissent. Furthermore, looking at impulse response functions during high and low dissent regimes, we see that they are similar and insignificant for most outcome variables, while showing diverging and significantly different interest rate paths, lending support to dissent being a monetary policy regime indicator. In conclusion, the insight that transparent dissent in a central bank has short run effects, but do not seem to affect longer term outcomes suggests that a central bank conducting transparent monetary policy does not trade off the effectiveness of forward guidance against the degree of transparency about its internal decision-making process.

A Appendix

A.1 Details on parsing and data extraction

To extract text data from the transcripts, I use pdfplumber. I then use regular expressions to extract each governor statement.

A.2 LLM and classification

A.2.1 Prompts

Model used: llama-3.3-70b-versatile

System prompt: You are a macroeconomist specializing in monetary policy and central bank communication analysis. You classify central bank statements at the sentence level.

Instruction:

Analyse the TARGET SENTENCE using the PREVIOUS SENTENCE as context (if provided). Also assign a probability - the confidence that the assigned score is correct. For each category, assign the required integer score and provide a short justification. If no previous sentence is provided, classify based on the target sentence alone. If unclear, assign score 0 and probability 0.6. Keep motivations concise (maximum 2 sentences). Do not restate either the target or the previous sentence. Base the classification strictly on explicit content. Do not infer policy intentions unless clearly implied. Return only valid JSON. Do not include any text before or after the JSON.

Definitions:

1. Economic Sentiment:

-1 = Negative economic outlook

0 = Neutral

1 = Positive economic outlook

2. Emotion:

0 = Objective, technical, neutral tone

1 = Emotionally charged or expressive language

3. Uncertainty:

0 = No explicit uncertainty expressed

1 = Explicit uncertainty about economic conditions or outlook

4. Hawkishness:

1 = Hawkish (if sentence provides information that suggests an increased likelihood of monetary tightening or implicitly implies such a stance. This includes statements

on increasing inflation or inflation pressure, increases in purchasing power or real incomes that signal excess demand or overheating risk, adverse supply shocks, low unemployment and higher wage growth, accelerating economic activity and a strong or even unsustainable money or credit growth. Furthermore, statements on public debt (a reduction in the deficit or avoidance of public debt is considered hawkish) or support for inflation-reducing economic policies should be classified as hawkish.)

-1 = Dovish (if a sentence provides information that suggests an increased likelihood of monetary easing or implicitly implies such a stance. This includes statements on decreasing inflation or easing inflation pressure, decreases in purchasing power or real incomes that signal weak demand or slack in the economy, favorable supply shocks, rising unemployment and lower wage growth, decelerating economic activity and weak or contracting money or credit growth. Statements that explicitly prioritize or express preference for supporting economic growth, employment, or output over price stability should also be classified as dovish. Furthermore, statements on public debt (an increase in the deficit or acceptance of higher public debt is considered dovish) or support for debt-financed government spending and other inflation-tolerant or growth-stimulating economic policies should be classified as dovish.)

0 = Neutral (descriptive statements without economic reference or statements on economic conditions without implications for monetary policy)

Output strictly valid JSON: {"sentiment":

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{ "motivation": string, "score":int },
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"emotion":

```
{ "motivation": string, "score":int },
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"uncertainty":

```
{ "motivation": string, "score":int },
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"hawkishness":

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{ "motivation": string, "score":int } }
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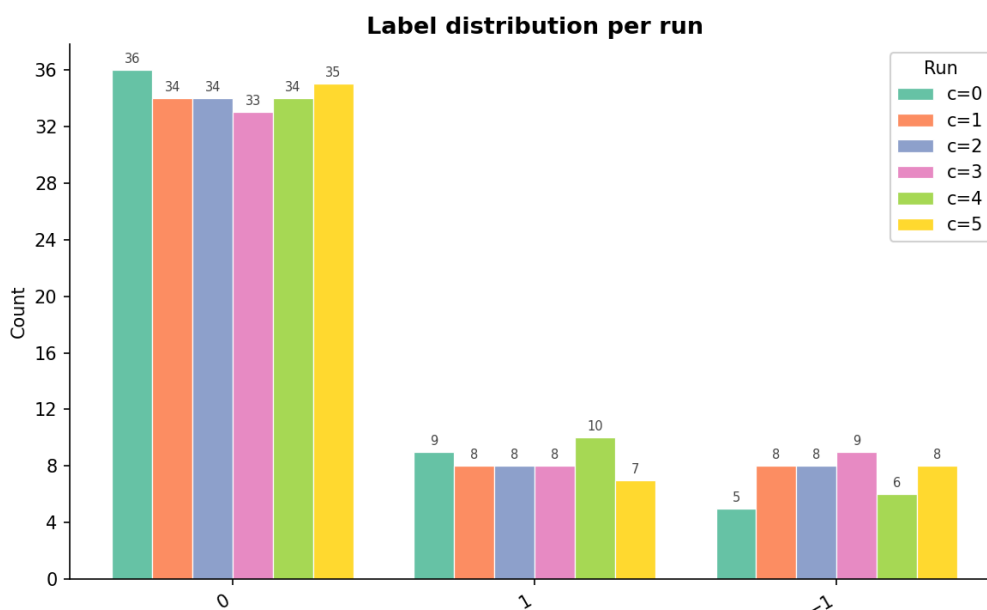
A.2.2 Details on labelling

A sentence is defined as...	...if it...
hawkish	• indicates increased likelihood of monetary tightening or mentions rising inflation or inflationary pressures • references low unemployment, accelerating wage growth or signals excess demand • describes accelerating economic activity, unsustainable money growth or notes adverse supply shocks • encourages deficit reduction, public debt avoidance or in some way endorses inflation-reducing economic policies.
dovish	• indicates likelihood of monetary loosening or mentions falling inflation or declining inflationary pressure • references concern about rising unemployment or describes weak or slowing economic activity • signals insufficient demand and supports debt-financed government spending.
neutral	• gives descriptive statements without economic reference or economic statements without implications for monetary policy.

Table 3: Details on sentence classification

A.2.3 Validation

Here I will show some results on how the LLM classification overlaps, depending on the number of context sentences. First, I classify the same 50 sentences with between 0 and 5 sentences of context to study the performance of the LLM.



I also perform a simple BOW measure of hawkishness to compare the performance of the LLM to this more

simple measure. I plot the analog to Figure 1 but also including the indices using a bag of words using the following two bags:

- **Hawkish words:** inflation, kpif, konsumentpris, producentpris, energipris, oljepris, råvarupris, utbudsstörning, kron, växelkurs
- **Dovish words:** tillväxt, resursutnyttjande, sysselsättning, konjunktur, finansiella, bnp, skuldsättning, bostadsmarknad, arbets, real

I then calculate a relative count score as

$$R_{gt} = \frac{\#hawk - \#dove}{\#hawk + \#dove}$$

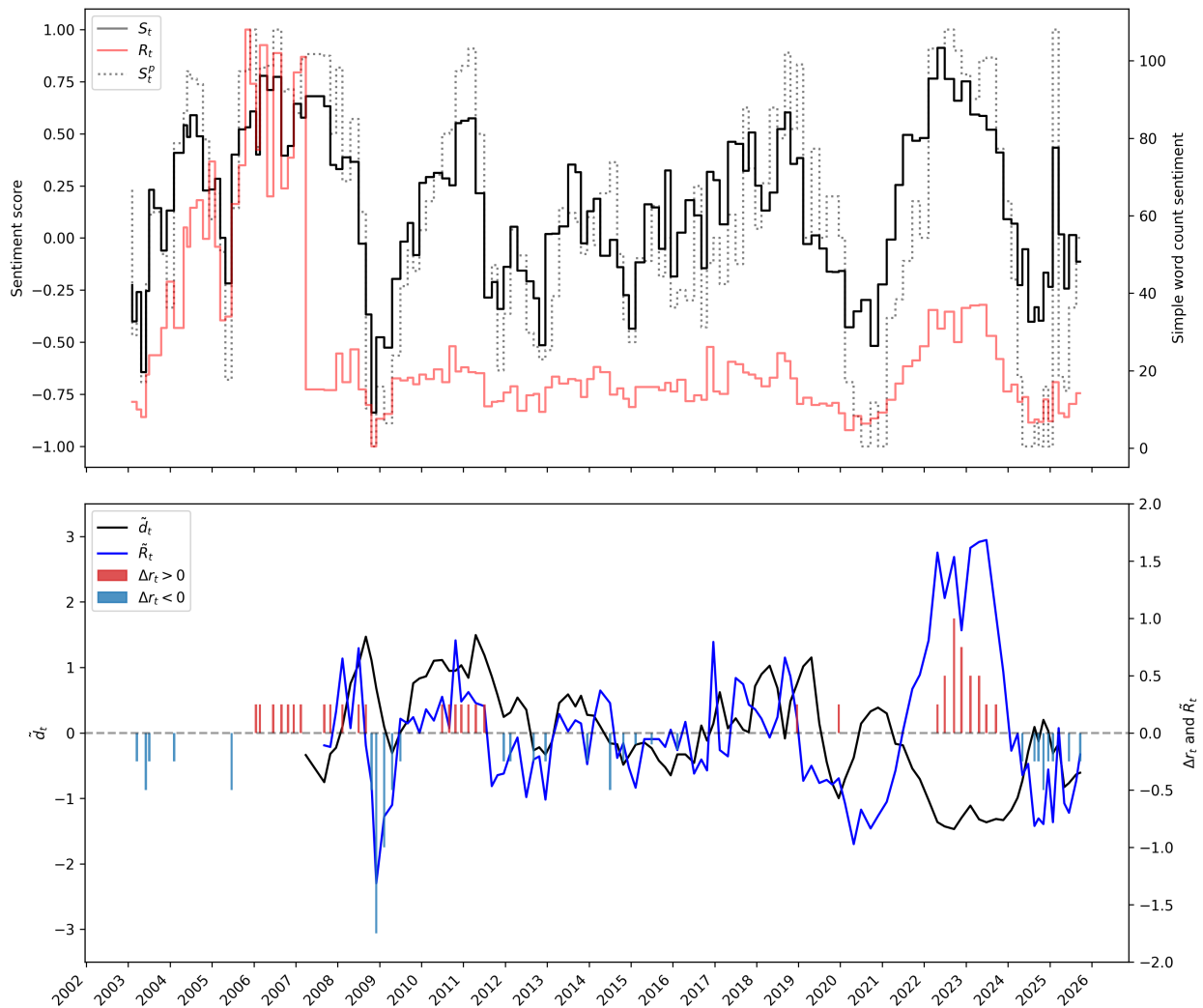


Figure 4: A reproduction of Figure 1 but also with simple BOW-index

What I see in Figure 4 is that the simple BOW algorithm yields little variation over time, and wrongly identifies the period in 2022-2024 as a period of high dissent. However, for the sentiment score I see only a tiny level of covariation and our $\rho \approx 0.15$. The major macroeconomic events in 2006-2009 and 2021-2024 are

somewhat picked up also by the simple bow-index.

A.3 Summary statistics

Table 4: Summary statistics collapsed by date

(1)					
	No. obs.	Average	S.d.	Min	Max
S_t	140	0.00	1.00	-2.71	2.15
S_t^p	140	0.00	1.00	-1.89	1.65
$\tilde{D}_t$	115	-0.00	1.00	-3.75	2.40
Δr_t	140	-0.01	0.27	-1.75	1.00
r_t	140	1.39	1.50	-0.50	4.75
v_t^{min}	48	-0.15	0.34	-0.75	0.75
D_t	48	-0.92	1.47	-3.00	3.00
Governor absent	3	1.00	0.00	1.00	1.00
α_{gt}	8	2.12	0.83	1.00	3.00
ι_{gt}	46	1.87	0.58	1.00	3.00

Table 5: Summary statistics by governor over time

Governor	Start	End	mean(S_t)	sd(S_t)	mean(length)	sd(length)	sum($ D_{it} $)	No. votes
Henry Ohlsson	2015-02	2023-02	0.317	0.500	62.391	19.470	7	46
Lars Nyberg	2007-09	2011-12	0.248	0.446	71.074	23.452	2	27
Svante Öberg	2007-09	2011-12	0.228	0.512	60.593	17.178	1	27
Stefan Ingves	2007-09	2022-11	0.213	0.422	76.852	24.292	1	90
Kerstin af Jochnick	2012-02	2019-09	0.188	0.337	54.156	12.708	1	45
Cecilia Skingsley	2013-07	2022-06	0.107	0.426	47.604	10.569	0	53
Anna Breman	2019-12	2025-09	0.100	0.473	67.743	14.912	2	35
Erik Thedéen	2023-02	2025-09	0.089	0.316	65.421	19.314	0	19
Per Jansson	2012-02	2025-09	0.089	0.398	68.852	17.317	2	82
Aino Bunge	2023-02	2025-09	0.059	0.459	54.684	11.392	0	19
Martin Flodén	2013-07	2024-05	0.037	0.369	59.651	16.086	7	63
Barbro Wickman-Parak	2007-09	2013-04	0.022	0.512	83.800	30.629	1	35
Anna Seim	2024-06	2025-09	-0.079	0.420	39.909	8.666	1	11
Irma Rosenberg	2007-09	2008-12	-0.136	0.494	58.333	31.093	2	9
Karolina Ekholm	2009-04	2014-09	-0.170	0.259	67.212	23.738	18	33
Lars E.O. Svensson	2007-09	2013-04	-0.202	0.427	124.118	67.728	23	34

Notes: The governor specific data set begins in 2007-09. Some governors have this as a start date because they were already part of the board when the data became transparent.

A.4 Governor specific plots

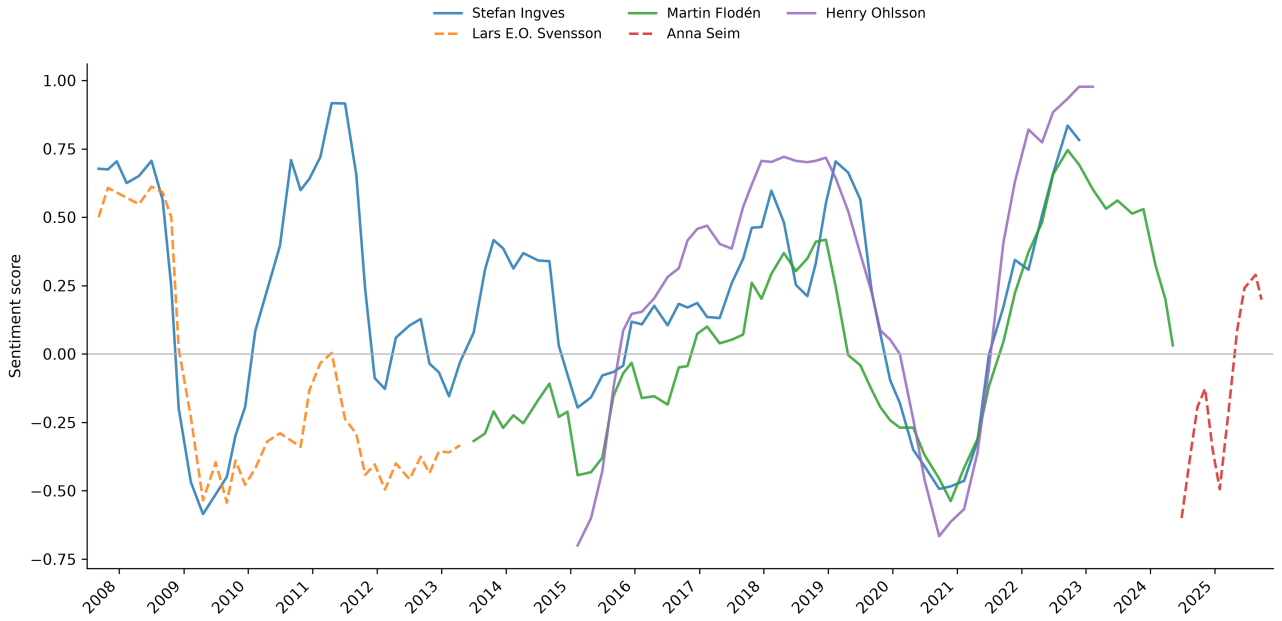


Figure 5: Governor specific sentiment scores over time.

A.5 Measures of dissent

I define three variables that capture these three types of dissent:

$$\text{Vote dissent: } \nu_{gt} := v_{gt} - \Delta r_t$$

Interest path dissent: $\iota_{gt} \in \{0, 1\}$, indicator whether governor g dissented to interest rate path at time t

Asset purchases dissent: $\alpha_{gt} \in \{0, 1\}$ indicator whether governor g dissented to asset purchases at time t

where v_{gt} is the vote of governor g and Δr_t is the repo rate change decided by the board, all in time t .

Table 6: Correlation of different dissent variables

(1)									
	ι_{gt}	α_{gt}	ν_{gt}	d_{gt}	$ D_{gt} $	$ d_{gt} $	$ \nu_{gt} $	S_{gt}	δ_{gt}
ι_{gt}	1.000								
α_{gt}	-0.050	1.000							
ν_{gt}	-0.536***	0.028	1.000						
d_{gt}	-0.260***	0.031	0.356***	1.000					
$ D_{gt} $	0.798***	-0.048	-0.583***	-0.287***	1.000				
$ d_{gt} $	0.310***	0.061*	-0.206***	-0.027	0.230***	1.000			
$ \nu_{gt} $	0.779***	-0.046	-0.679***	-0.290***	0.958***	0.224***	1.000		
S_{gt}	-0.160***	0.012	0.255***	0.550***	-0.171***	-0.134***	-0.181***	1.000	
δ_{gt}	-0.383***	0.067	0.453***	0.982***	-0.387***	-0.092	-0.389***	0.664***	1.000

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

As we can see, dissenting on the interest path is strongly correlated to dissent in voting and also to dissent in sentiment. The negative sign on the correlations between ι_{gt} and ν_{gt} and d_{gt} stems from the fact that

throughout the time series, dissent has occurred more often in the dovish direction ($\sim 2007-2014$). As for dissent in the asset purchasing dimension, α_{gt} , we see that it correlates very weakly with $|d_{gt}|$ and not with any of the other variables. Another measure of dissent that I study using the sentiment score is the mean difference in the sentiment score of the majority and the minority when there is vote dissent. We overlay this with the mean difference in voting on the interest rate of the majority vs. the minority ($\frac{1}{|G_t|} \sum_g \nu_{gt} : \nu_{gt} := v_{gt} - \Delta r_t$) and plot this in Figure 6 to highlight the way in which vote dissent and sentiment dissent covary.

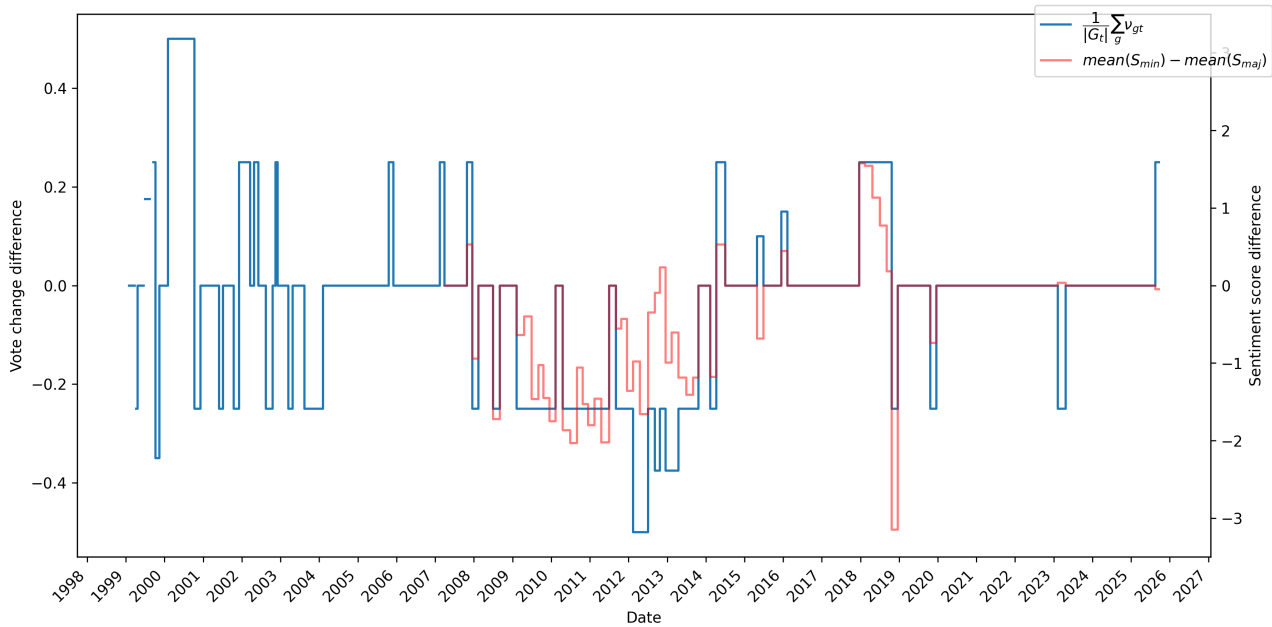


Figure 6: Intensity in voting dissent and sentiment deviation of dissenting minority

In the figure, we can see that there are two periods with consistent dissent. One dissenting dovish period beginning in 2009 where one to two members of the executive board consistently disagreed on policy and deviated, both in speech and in voting in a dovish direction. The peak dissent deviation with the majority of the board in this first period occurred in January of 2011. After that, there was some moderation in the deviation of the minority where eventually the prominent dissenters a bit later left the board. Following this, there is from 2018 and onwards a dissenting hawkish period which ends with the data set's biggest outlier in the dovishness dimension.⁸ Before we have individual governor sentiment data, we also observe dissent but only in the voting. As I see, when data is available for the dissenting minority's sentiment, it tracks voting dissent quite well. There are only a few notable exceptions when dissent in voting is in the positive direction and sentiment deviates in the other direction, for example the 15 July 2015⁹.

To put the governor dissent variables into context, I provide some correlations below in Table 6.

⁸Governor Per Jansson expresses great concerns for too weak inflation and wants to keep the interest rate unchanged at -0.5 %.

⁹The governor in that case, Henry Ohlsson, still saw it as necessary to stimulate the economy but expressed a distaste for letting the interest rate get more negative.

A.6 Alternate regressions

A.6.1 Interest rate dependence with different weights

Table 7: Weighted by number of sentences spoken

	(1)	(2)	(3)	(4)	(5)
	Δr_t	Δr_t	Δr_t	Δr_t	Δr_t
S_t	0.600*** (0.157)	0.545*** (0.151)	0.433** (0.211)	0.427* (0.232)	0.371* (0.199)
Inflation controls		0.063** (0.025)	0.060** (0.024)	0.087*** (0.026)	0.048* (0.025)
Unemployment controls		-0.033 (0.066)	-0.020 (0.070)	0.045 (0.072)	-0.020 (0.065)
S_t^p			0.086 (0.069)	0.069 (0.076)	0.044 (0.079)
Δ_t				-0.148** (0.062)	-0.116** (0.056)
$\Delta r_{g,t-1}$					0.372*** (0.113)
Constant	-0.085*** (0.031)	0.047 (0.500)	-0.035 (0.522)	-0.580 (0.546)	0.014 (0.501)
R-squared	0.602	0.629	0.633	0.652	0.704
F-stat	14.596	7.233	8.158	6.533	7.879
No. obs.	140.000	140.000	140.000	113.000	113.000

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 8: Weighted by dissent (0) or not (1) - majority "control"

	(1) Δr_t	(2) Δr_t	(3) Δr_t	(4) Δr_t	(5) Δr_t
S_t	0.569*** (0.134)	0.523*** (0.128)	0.483*** (0.169)	0.533** (0.206)	0.489*** (0.179)
Inflation controls		0.060** (0.024)	0.058** (0.024)	0.082*** (0.026)	0.038 (0.027)
Unemployment controls		-0.034 (0.064)	-0.028 (0.067)	0.046 (0.094)	-0.020 (0.077)
S_t^p			0.035 (0.062)	0.010 (0.076)	-0.008 (0.081)
Δ_t				-0.134 (0.095)	-0.083 (0.088)
$\Delta r_{g,t-1}$					0.383*** (0.114)
Constant	-0.102*** (0.032)	0.043 (0.481)	0.010 (0.500)	-0.602 (0.712)	0.002 (0.600)
R-squared	0.634	0.658	0.658	0.667	0.721
F-stat	18.072	8.363	7.550	5.593	7.755
No. obs.	140.000	140.000	140.000	97.000	97.000

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 9: Keeping only chair of EB

	(1) Δr_t	(2) Δr_t	(3) Δr_t
S_t	0.427*** (0.142)	0.371*** (0.133)	0.183 (0.116)
Inflation controls		0.084*** (0.025)	0.075*** (0.022)
Unemployment controls		-0.003 (0.097)	0.047 (0.091)
S_t^p			0.217*** (0.067)
Constant	-0.089** (0.043)	-0.228 (0.730)	-0.563 (0.685)
R-squared	0.530	0.572	0.609
F-stat	8.997	7.194	7.497
No. obs.	107.000	107.000	107.000

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

A.6.2 Governor effect on interest rates

Now we will continue by looking at governor specific effects. We want to know if there are governors that are especially influential when it comes to the messaging conveyed in the press releases at the time of decisions. We also want to know if there are individual governors that deviate and by deviating from the others can

influence the decision in any direction. We will study this by first looking at the following regression:

$$S_t^p = \alpha_0 + \sum_{g \in G} \beta_g S_{gt} + \Gamma X_t + \varepsilon_t \quad (9)$$

S_t^p is the sentiment of the preceding press release in time t , S_{gt} is the sentiment score of governor g in time t . I expand the regression by including controls, X_t for inflation, unemployment and time fixed effects (year). We can see the results from this in Table 10. Almost all governors' statements are strongly correlated with the message that is conveyed in the preceding PR's. However the size of the correlation varies quite a lot. The big dissenters (LEO and KE) seem not to be less visible in the PR's.

Table 10: Results from regression (9)

	(1) S_t^p	(2) S_t^p
Aino Bunge $\times S_t$	0.696*** (0.242)	0.761*** (0.242)
Anna Breman $\times S_t$	0.619*** (0.148)	0.675*** (0.151)
Anna Seim $\times S_t$	0.547* (0.300)	0.477 (0.319)
Barbro Wickman-Parak $\times S_t$	0.676*** (0.095)	0.699*** (0.097)
Cecilia Skingsley $\times S_t$	0.304*** (0.074)	0.304*** (0.076)
Erik Thedéen $\times S_t$	0.921*** (0.296)	1.047*** (0.306)
Henry Ohlsson $\times S_t$	0.262*** (0.056)	0.310*** (0.066)
Irma Rosenberg $\times S_t$	0.774*** (0.153)	0.955*** (0.280)
Karolina Ekholm $\times S_t$	0.548*** (0.172)	0.989*** (0.188)
Kerstin af Jochnick $\times S_t$	0.337*** (0.089)	0.400*** (0.087)
Lars E.O. Svensson $\times S_t$	0.363*** (0.127)	0.479*** (0.150)
Lars Nyberg $\times S_t$	0.728*** (0.118)	0.917*** (0.099)
Martin Flodén $\times S_t$	0.421*** (0.082)	0.407*** (0.076)
Per Jansson $\times S_t$	0.382*** (0.102)	0.413*** (0.108)
Stefan Ingves $\times S_t$	0.498*** (0.064)	0.604*** (0.074)
Svante Öberg $\times S_t$	0.695*** (0.093)	0.812*** (0.088)
Inflation controls	0.092** (0.043)	0.085* (0.043)
Unemployment controls	-0.197* (0.105)	-0.200* (0.101)
Constant	1.282 (0.851)	1.316 (0.827)
Governor FE	No	Yes
R-squared	0.719	0.739
F-stat	7.506	10.676
No. obs.	624.000	624.000

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

A.7 Variance analysis

To further analyze whether the results on reduced size effects on short term OIS rates, we look to data on volatility and difference between the highest and lowest quote price around the releases of .

$$var(Y_t) = \alpha_0 + \beta_0 \tilde{d}_t + \varepsilon_t \quad (10)$$

Table 11: Results from regression 10

	(1) 1M	(2) 1M	(3) 3M	(4) 3M	(5) OMXSPI	(6) OMXSPI
Δ_t	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)	0.032 (0.052)	0.015 (0.047)
r_t	0.000 (0.000)	0.000 (0.000)	0.001 (0.001)	0.001 (0.001)	0.272 (0.241)	0.017 (0.100)
VIX		0.000 (0.000)		0.001 (0.000)		-0.027 (0.061)
Constant	-0.000 (0.000)	-0.000 (0.000)	-0.001 (0.001)	-0.001 (0.001)	-0.401 (0.286)	-0.086 (0.097)
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
R-squared	0.523	0.524	0.236	0.372	0.173	0.059
Mean outcome size						
F-stat	0.899	0.678	0.448	0.620	0.654	0.121
No. obs.	39.000	39.000	56.000	56.000	109.000	108.000

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

$$maxmin(Y_t) = \alpha_0 + \beta_0 \tilde{d}_t + \varepsilon_t \quad (11)$$

Table 12: Results from regression 11

	(1)	(2)	(3)	(4)	(5)	(6)
	1M	1M	3M	3M	OMXSPI	OMXSPI
Δ_t	0.002 (0.003)	0.003 (0.003)	0.020 (0.018)	0.011 (0.011)	0.029 (0.092)	0.002 (0.089)
r_t	0.021 (0.019)	0.046 (0.034)	0.043 (0.034)	0.052 (0.033)	0.217 (0.320)	-0.008 (0.262)
VIX		0.004 (0.008)		0.031 (0.020)		0.066 (0.122)
Constant	-0.013 (0.019)	-0.041 (0.037)	-0.036 (0.043)	-0.056 (0.046)	-0.497 (0.368)	-0.218 (0.282)
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
R-squared	0.361	0.415	0.260	0.388	0.136	0.112
Mean outcome size						
F-stat	0.742	0.625	0.816	0.996	0.236	0.128
No. obs.	61.000	59.000	70.000	68.000	109.000	108.000

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

A.8 Robustness checks of head line result

Table 13: Robustness check of m1 result looking at sign restricted shocks

	(1)	(2)
	3m	3m
$\text{lm}_{dissent_d} \text{direction}$	-0.908 (0.544)	0.072 (0.289)
Constant	0.962*** (0.353)	-0.210 (0.203)
R-squared	0.253	0.199
F-stat	2.786	0.063
No. obs.	61.000	73.000

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 14: Robustness check of m1 result

	(1)	(2)	(3)	(4)	(5)	(6)
	3m	3m	3m	3m	3m	3m
$\text{lm}_{dissent_d} \text{direction}$	-0.500 (0.323)	-0.615* (0.360)	-0.225 (0.200)	-1.349** (0.539)	-0.346 (0.470)	-0.791* (0.402)
Constant	1.158*** (0.148)	1.178*** (0.200)	1.166*** (0.179)	1.080** (0.449)	1.026*** (0.372)	1.460*** (0.226)
Sample	Drop LEO	Drop t/b 5 %	No vote dissent	With vote dissent	2007-2015	2015-2025
R-squared	0.297	0.199	0.224	0.315	0.132	0.316
Mean outcome size						
F-stat	2.388	2.926	1.270	6.268	0.542	3.873
No. obs.	106.000	104.000	63.000	36.000	45.000	64.000

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

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